

GCE O-LEVEL MATHEMATICS PRACTICE PAPER 5

Statistics & Probability

Duration: 2.5 hours Total Marks: 100
Name: _____ Date: _____

SECTION A: Short Answer Questions (40 marks)

1. Find mean, median, mode of: 3, 5, 7, 5, 9, 5, 11 [3 marks]

2. Calculate standard deviation of: 2, 4, 6, 8, 10 [4 marks]

3. Two dice rolled. $P(\text{sum} = 7) = ?$ [2 marks]

4. If $P(A) = 0.3$ and $P(B) = 0.4$, and independent, find $P(A \cap B)$ [2 marks]

5. Probability of selecting red marble from bag with 5 red, 3 blue, 2 green? [2 marks]

6. Find z-score if $x = 75$, mean = 80, std dev = 5 [2 marks]

7. Arrange in ascending order by probability: A has $P=0.5$, B has $P=0.25$, C has $P=0.75$ [2 marks]

8. In a class of 30, 12 like football, 18 like cricket. If 5 like both, how many like at least one? [3 marks]

SECTION B: Long Answer Questions (60 marks)

9. The heights of 100 students have mean 170 cm, std dev 10 cm. Assuming normal distribution, find how many students are between 160-180 cm. (5 marks)

10. A coin is flipped 4 times. Find probability of exactly 2 heads. (4 marks)

11. A committee of 3 is chosen from 8 people. How many ways can this be done? (3 marks)

12. Two bags: Bag A has 3 red, 2 blue. Bag B has 2 red, 3 blue. Draw one marble from each. P(both red)? (4 marks)

13. Permutations: How many arrangements of word 'STATISTICS'? (5 marks)

14. A frequency table shows: 10-20 (5), 20-30 (8), 30-40 (12), 40-50 (5). Find median class and estimate median. (6 marks)

SOLUTIONS

SECTION A SOLUTIONS

Q1: Mean, median, mode

Mean = $(3+5+7+5+9+5+11)/7 = 45/7 \approx 6.4$, Median = 5 (middle value when sorted), Mode = 5 (appears 3 times)

Q2: Standard deviation

Mean = 6, Variance = $[(2-6)^2 + (4-6)^2 + (6-6)^2 + (8-6)^2 + (10-6)^2]/5 = (16+4+0+4+16)/5 = 8$, Std Dev = $\sqrt{8} = 2\sqrt{2} \approx 2.83$

Q3: P(sum = 7)

Favorable: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) = 6, Total = 36, $P = 6/36 = 1/6$

Q4: P(A $\cap$ B) independent

$P(A \cap B) = P(A) \times P(B) = 0.3 \times 0.4 = 0.12$

Q5: P(red marble)

Total = $5 + 3 + 2 = 10$, $P(\text{red}) = 5/10 = 0.5$

Q6: Z-score

$z = (x - \text{mean})/\text{std dev} = (75 - 80)/5 = -1$

Q7: Arrange by probability

B (0.25), A (0.5), C (0.75)

Q8: Venn diagram

Football only = $12 - 5 = 7$, Cricket only = $18 - 5 = 13$, Both = 5, Neither = $30 - 25 = 5$, At least one = $7 + 5 + 13 = 25$

SECTION B SOLUTIONS

Q9: Normal distribution

Between 160-180 is ± 1 std dev from mean, approximately 68% = $0.68 \times 100 = 68$ students

Q10: Binomial probability

$P(X = 2) = C(4,2) \times (0.5)^2 \times (0.5)^2 = 6 \times 0.0625 = 0.375$

Q11: Combinations

$C(8,3) = 8!/(3!5!) = (8 \times 7 \times 6)/(3 \times 2 \times 1) = 56$ ways

Q12: P(both red)

$P(A \text{ red}) = 3/5$, $P(B \text{ red}) = 2/5$, $P(\text{both}) = (3/5) \times (2/5) = 6/25$

Q13: Permutations of STATISTICS

S appears 3 times, T appears 2 times, A, I, C appear 1 time each, $P = 10!/(3! \times 2!) = 3,628,800/12 = 302,400$

Q14: Median class and estimate

Median class is 30-40 (cumulative frequency reaches 50th item), Estimated median $\approx 30 + (5/12) \times 10 = 34.2$